Mar & Kuku Production Studio

Verified Architecture, Corrected Claims & Operator Testing Guide (Milestones 1–9) • October 2026

Verified System Integrity & Honest Operational Distinctions:
• **Production Library:** `279 / 279` files in `/Users/yonassayfu/Desktop/DoroNesh_2D_Animation/` strictly read-only and byte-identical.
• **Automated Test Suite:** `62 passed` across all components (`.venv/bin/pytest -q`).
• **M0 CH01 Technical Proof:** `mk-studio verify-ch01` passed (`exit_code=0` , bit-identical audio PCM, 4.00s stillness hold).
• **Dynamic Script-Driven Execution:** Synthesizes real dialogue and holds dynamically parsed from the active Markdown script without Python edits. Verified across distinct scripts.
• **Renderer Distinction:** Distinguishes native Palmier timeline assembly (verified on disk) from deterministic FFmpeg MP4 previews.
• **YouTube Status:** Accurately distinguishes upload packaging & browser staging from finalized channel publication.

1. 3-Agent Live Consultation (Steps 1–5)

Step	Agent & URL	Function & Operational Distinction	Status
1	**ChatGPT Web** (Master) `https://chatgpt.com/`	Drafts educational arc, scene outline, character dialogue, and specialist questions. Checkpointed in `01_chatgpt_master_plan.json` .	COMPLETED
2	**Google Gemini** (Reviewer) `https://gemini.google.com/app`	Extracted directly from live DOM (`model-response` / `message-content`). Reviews preschool 1-to-3 counting clarity, hard pose swaps, and response hold. Non-empty response verified.	COMPLETED
3	**Manus Web** (Specialist) `https://manus.im/app/`	Authenticated user session (Guangut Semera). Evaluates speech pacing (110–120 WPM), 12-frame counting pauses, and hold window timing.	COMPLETED
4	**ChatGPT Synthesis** `https://chatgpt.com/`	Synthesizes master plan, reviewer feedback, and specialist timing into approved script (`v001`).	COMPLETED
5	**Follow-Up Correction** ChatGPT + Gemini	Applies owner revision into `v002` without re-running steps 1–4. Confirmed by Gemini.	COMPLETED

2. Production & Delivery Pipeline (Steps 6–9)

Step	Stage	Artifacts & Verifications	Renderer / Mode
6	**Asset Generation & Intake**	Dynamic voiceover synthesized from parsed script cues (Samantha/Karen); high-res visual props; 1280x720 thumbnail; calibrated `ChapterSpec` and `AssetManifest` .	macOS Speech + FFmpeg (Local) / Browser Intake
7	**Palmier Timeline Editing**	Native Palmier project bundle compiled: 5 tracks, 20 clips, 6 markers. Verified on-disk structure and close/reopen reconciliation.	Palmier Pro Native Bundle (`.palmier`)
8	**Final MP4 Export & QA**	1080p30 900-frame video. Computer vision frame QA passed: Bob RMS = 28.39, Stillness RMS = 0.57 (frames 501..620). Full decode 0 errors.	Deterministic FFmpeg Preview (Verified against Palmier Plan)

Step	Stage	Artifacts & Verifications	Renderer / Mode
9	**YouTube Delivery**	Prepares complete metadata package (Title, Description, Tags, Category 27, Made for Kids). Stages package in managed browser session; supports automated file upload.	YouTube Studio Browser Automation (Private Mode)

3. Episode 002 Approved Nine-Chapter Architecture (CH01–CH09)

In accordance with `docs/MK_Production_Studio_Requirements_v001.md` and `E02_MASTER_SCRIPT_v001.md`, Episode 002 is officially defined as a **9-chapter journey**:

Chapter	Title & Educational Content	Key Assets & Actions	Hold Window
CH01	Welcome to the Colour Garden	Meet Mar & Kuku; Scandinavian garden; empty baskets revealed.	4s response hold
CH02	Meet Red, Yellow and Blue	Teaching Group 1 flower introductions; petal emphasis.	4s response hold
CH03	Meet Green, Orange and Purple	Teaching Group 2 flower introductions; petal emphasis.	4s response hold
CH04	First-Group Basket Matching	Match Red, Yellow, Blue flowers into neutral baskets.	4s response hold
CH05	Second-Group Basket Matching	Match Green, Orange, Purple flowers into neutral baskets.	4s response hold
CH06	Three Colour-Detective Rounds	Interactive visual search rounds; finger-pointing cues.	4s response hold
CH07	Gentle Original Colour Song	Acoustic preschool melody reinforcing the six colours.	Musical breathing
CH08	Recall Red, Yellow, Orange	Active memory retrieval challenge with preschool pacing.	4s response hold
CH09	Picnic Celebration & Goodbye	Complete picnic reveal, celebration dance, and host goodbye wave.	Closing wave

4. Step-by-Step Operator Instructions

Action A: Start the Studio Server

```
.venv/bin/mk-studio studio --port 8765
# Open in browser: http://127.0.0.1:8765/
```

Action B: Execute Dynamic 3-Agent Script Workflow

Click **1-4: 3-Agent Script** or send:

```
Research the next Mar and Kuku episode, consult Gemini and Manus, and save the proposed script.
```

The system checkpoints progress in `output_manifest.json`. If stopped, clicking **Resume** continues from the next uncompleted step without re-submitting earlier steps.

Action C: Run Dynamic Production Pipeline

Click **Full Pipeline (6-9)**. The system:

1. Parses the active approved script into dynamic dialogue cues and stillness hold boundaries.
2. Synthesizes master audio and generates visual teaching props.
3. Compiles the native Palmier project (`.palmier`) and reconciles timeline clips on disk.
4. Renders the 1080p30 video and runs computer vision frame QA.

5. Stages the package for YouTube Studio in the managed browser.

5. Architectural Truth & Honest Status Summary

- **Script Parsing:** The voiceover is not hardcoded. Spoken dialogue is parsed directly from the selected script file.
- **Asset Generation:** Local macOS Speech and procedurally drawn props are clearly labeled as fast deterministic options. External browser-generated assets are validated via `BrowserAssetGenerator.collect`.
- **Palmier Workflow:** Native `.palmier` project bundles on disk are verified via `verify_native_palmier_project`. MP4 rendering is labeled as deterministic FFmpeg preview.
- **YouTube Status:** Step 9 stages metadata and verifies dashboard readiness; full upload execution is protected by private visibility and owner approval.